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AI Research Agent : A Structured Multi-Stage LLM System for Automated Research Report Generation

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AI Research Project

LLM-Based Research Automation System

1. Abstract

Large Language Models (LLMs) are capable of generating fluent responses to complex questions, but they do not inherently perform structured research or evidence validation. Standard prompt-based usage often produces answers that lack multi-source grounding, cross-source agreement analysis, and explicit handling of factual conflicts. This project presents the **AI Research Agent**, a modular research automation system that treats LLMs as controlled reasoning components within a structured pipeline rather than as a single end-to-end generator.

The system performs research through a multi-stage workflow that includes research planning, iterative evidence discovery, dense summarization, cross-source agreement and conflict analysis, evidence-grounded report synthesis, and plan-aware evaluation. A persistent vector memory enables knowledge reuse, while credibility scoring and agreement-based weighting improve evidence reliability. Conflict detection and resolution ensure that contradictory claims are explicitly handled rather than silently merged.

The resulting system behaves more like an automated research assistant than a chatbot, producing structured academic-style reports with sentence-level citations and traceable evidence. Experimental observations show that iterative discovery improves coverage, agreement scoring strengthens reliable evidence, and citation-constrained synthesis reduces hallucination. This work demonstrates a full research reasoning architecture for LLM-based systems, emphasizing structure, traceability, and multi-source validation.

2. System Architecture

This section presents the overall architecture of the AI Research Agent, highlighting how structured reasoning stages are organized into a modular research pipeline. The system is designed to treat Large Language Models (LLMs) as controlled reasoning components rather than as a single end-to-end generator.

The architecture enables the system to conduct structured, multi-source research through separated cognitive stages, ensuring traceability, evidence grounding, and reduced hallucination risk.

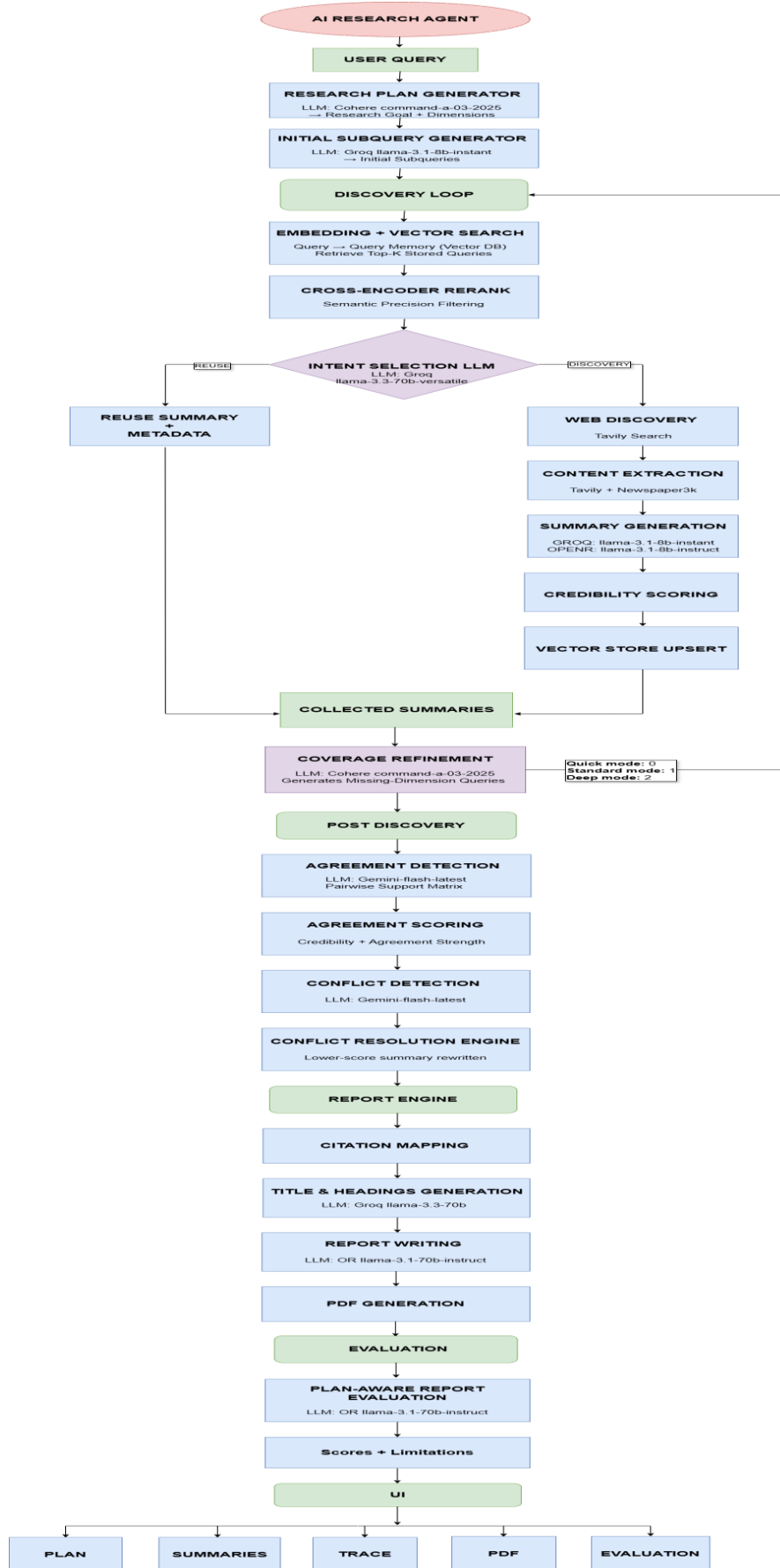


Figure 1: System Architecture of the AI Research Agent

Architecture Overview

The AI Research Agent follows a disciplined research pipeline inspired by human research methodology:

Planning → Discovery → Validation → Synthesis → Evaluation

Each stage represents a distinct reasoning function, and together they form a closed research loop that transforms a user query into a grounded academic-style report.

Architectural Layers

1. Research Planning Layer

This layer converts the raw user query into a structured research scope.

- Generates research dimensions
- Produces broad subqueries
- Defines the boundaries of investigation

This stage ensures the system plans *what to research* before retrieving information.

2. Iterative Discovery Engine

This is the exploration core of the system.

- Searches vector memory for reusable knowledge
- Performs web retrieval when necessary
- Summarizes retrieved evidence
- Stores summaries in persistent memory
- Refines coverage through iterative querying

This layer is responsible for *finding and expanding evidence*.

3. Cross-Source Reasoning & Validation

This layer performs structured evidence analysis.

- Detects agreement between sources
- Identifies factual conflicts
- Assigns evidence strength scores
- Resolves contradictions

This ensures that evidence is **weighted and validated**, not simply aggregated.

4. Report Synthesis Engine

This layer converts validated evidence into a structured report.

- Organizes sections and headings
- Generates academic-style synthesis
- Enforces sentence-level citations
- Prevents unsupported claims

This stage transforms evidence into a readable, traceable research document.

5. Evaluation Engine

The final stage checks research quality.

- Verifies grounding in evidence
- Measures coverage completeness
- Assesses structural quality
- Evaluates citation correctness

This closes the reasoning loop and ensures alignment with the research plan.

Architectural Significance

The AI Research Agent differs from standard LLM applications in several important ways:

- LLMs are used as components, not as the system itself
- Research is iterative, not one-shot
- Evidence is validated through agreement and conflict analysis
- Memory enables knowledge reuse across research cycles
- Final outputs are citation-grounded and auditable

3. Research and Reasoning Workflow

This section explains the step-by-step reasoning process used by the AI Research Agent to conduct structured, multi-source research. While the architecture describes system components, this workflow describes how information flows through those components to produce a grounded research report.

The system follows a disciplined reasoning loop inspired by human research methodology:

Plan → Discover → Validate → Synthesize → Evaluate

3.1 Research Planning

The workflow begins by converting the user’s research question into a structured investigation plan.

At this stage, the system:

- Identifies key research dimensions
- Breaks the question into broad subqueries
- Defines the scope of investigation

This ensures that the system determines what needs to be researched before retrieving information, preventing unstructured or drifting exploration.

3.2 Iterative Evidence Discovery

This is the core exploration phase where the system gathers evidence.

For each generated subquery, the system performs the following:

1. **Memory Search**
The vector database is queried to find previously stored summaries related to the current research intent. This enables knowledge reuse and avoids redundant web searches.
2. **Semantic Reranking**
Retrieved candidates are refined using cross-encoder reranking to improve semantic precision.
3. **Intent Matching Decision**
A language model determines whether an existing summary already answers the same research intent.
 - If yes → The summary is reused.
 - If no → The system proceeds to web discovery.
4. **Web Discovery Pipeline**
When new information is required, the system:
 - Retrieves web sources
 - Extracts article content
 - Generates dense factual summaries
 - Scores source credibility
 - Stores validated summaries in memory

After each round, the system evaluates research coverage. If gaps remain, it generates additional targeted queries. This creates an iterative research expansion loop rather than a one-shot search.

3.3 Cross-Source Validation

Once summaries are collected, the system performs structured evidence reasoning.

This stage includes:

- **Agreement Detection**
Identifies when multiple sources support similar claims.
- **Agreement Scoring**
Strengthens evidence that has cross-source support.
- **Conflict Detection**
Identifies hard factual contradictions between summaries.
- **Conflict Resolution**
Lower-scoring conflicting claims are removed or rewritten.

This transforms raw summaries into trust-weighted evidence, ensuring that the final synthesis reflects the most reliable information.

3.4 Evidence-Grounded Report Synthesis

Validated summaries are then converted into a structured academic-style report.

The system:

- Organizes content into logical sections
- Synthesizes findings from multiple sources
- Enforces sentence-level citations
- Prevents unsupported claims

This ensures the report is not just fluent text, but a traceable synthesis grounded in collected evidence.

3.5 Plan-Aware Evaluation

In the final stage, the system evaluates its own output.

The evaluation checks whether:

- All planned research dimensions are covered

- Claims are supported by collected summaries
- The report structure is coherent
- Citations are correctly applied

This closes the reasoning loop and ensures alignment between the initial research plan and the final generated report.

4. Design Decisions and Limitations

This section discusses the key architectural and methodological decisions behind the AI Research Agent, followed by known system limitations. These design choices were made to improve reliability, evidence grounding, and reasoning transparency compared to standard single-prompt LLM systems.

4.1 Design Decisions

Multi-Model Orchestration Instead of a Single LLM

Rather than relying on one large model for all tasks, the system assigns different models to specialized roles such as planning, summarization, reasoning, synthesis, and evaluation. This approach balances reasoning depth, speed, and cost efficiency while improving robustness.

Structured Research Planning Before Retrieval

The system does not begin with web search immediately. Instead, it first generates research dimensions and subqueries. This ensures that information gathering is goal-driven and prevents scope drift.

Iterative Coverage Refinement

Research is treated as an iterative process. After initial discovery, the system evaluates coverage and generates follow-up queries for missing areas. This mimics how human researchers expand their investigation when gaps are identified.

Persistent Vector Memory

Collected summaries are stored in a vector database. This enables:

- Knowledge reuse
- Reduced redundant searches

- Faster future research cycles

This design transforms the system from stateless querying into a learning research assistant.

Agreement-Based Evidence Weighting

Evidence supported by multiple independent sources is given higher strength. This reduces the impact of isolated or weak claims and promotes trust-weighted synthesis.

Explicit Conflict Detection and Resolution

Instead of ignoring contradictions, the system actively detects factual conflicts and removes weaker claims. This prevents silent merging of inconsistent information.

Citation-Constrained Report Generation

Every claim in the final report must be grounded in collected summaries. Sentence-level citations enforce traceability and reduce hallucination risk.

Plan-Aware Evaluation

The system evaluates outputs based on how well they satisfy the original research plan, not just fluency. This aligns the final result with intended research goals.

4.2 Limitations

Despite structured design, the system has important constraints.

Dependence on Source Quality

The system's knowledge is derived from web sources. If sources are inaccurate, biased, or outdated, the final synthesis may reflect those weaknesses.

Probabilistic Nature of LLM Reasoning

Agreement detection, conflict detection, and evaluation rely on LLM judgments. While prompts are constrained, edge-case reasoning errors may still occur.

Memory Bias

Reusing stored summaries improves efficiency but may over-represent earlier topics or perspectives.

Sensitivity to Query Clarity

Poorly defined research questions can lead to suboptimal research plans and incomplete coverage.

Model-Based Evaluation

The evaluator is also an LLM. Although constrained to system outputs, evaluation reflects structured automated judgment rather than human peer review.

5. Output Demonstration

This section demonstrates how the AI Research Agent operates in practice, from user interaction to final report generation and evaluation. Instead of static screenshots, the system behavior is illustrated through the internal outputs produced across different interface tabs.

The process begins when a user provides a research query and selects a discovery mode. The system then executes the structured research pipeline and exposes intermediate reasoning artifacts, allowing transparency into how the final report is constructed.

5.1 Research Plan

After receiving the query “*Assess the multifaceted impact of generative AI on software engineering roles*”, the system first constructs a structured research plan.

Goal

Assess the multifaceted impact of generative AI on software engineering roles

Research Dimensions

- Job Task Transformation
- Skillset Evolution Requirements
- Employment Market Dynamics
- Productivity and Quality Metrics
- Ethical and Societal Implications
- Organizational Adoption Strategies

This stage defines the scope of investigation before any evidence retrieval begins, ensuring goal-driven research rather than unstructured search.

5.2 Trace of Research Execution

The trace tab provides a transparent view into the internal reasoning pipeline of the AI Research Agent. Rather than functioning as a black-box model, the system exposes each decision stage that leads from the research plan to the final synthesized report.

The trace demonstrates that research is conducted as a controlled, multi-stage reasoning process rather than a single retrieval or generation step.

Step 1 — Research Plan Initialization

The workflow begins with the structured research plan derived from the user's query. The goal and dimensions are logged as the guiding blueprint for all subsequent decisions. This ensures that every discovery action is aligned with predefined research objectives.

Step 2 — Discovery Iteration 1

The system generates initial subqueries targeting:

- Job task changes
- Workforce skills and employment trends

These subqueries serve as focused search intents rather than generic keywords.

Step 3 — Vector Memory Search

For each subquery, the vector database is searched for semantically similar prior research queries.

Purpose:

- Avoid redundant web searches
- Enable knowledge reuse
- Leverage previously validated evidence

Multiple candidate matches (VS_01–VS_10) are retrieved for each query.

Step 4 — Cross-Encoder Reranking

The retrieved candidates are then reranked using a cross-encoder model. This step improves semantic precision by performing pairwise comparison between the subquery and candidate intents.

This transforms approximate vector similarity into fine-grained semantic matching.

Step 5 — Intent Selection Decision

A language model determines whether any stored summary matches the same research intent.

Observed result:

- For both initial queries, **no reusable intent** was selected.

This decision triggers web-based discovery, demonstrating that the system does not reuse memory blindly.

Step 6 — Web Ingestion and Summary Generation

Since reusable evidence was unavailable, the system performs web retrieval:

- Sources are collected
- Article content is extracted
- Dense factual summaries are generated
- Credibility scoring is applied
- Summaries are stored in memory

Two independent providers (Groq and OpenRouter) were used for summarization, increasing robustness.

Step 7 — Coverage Refinement (Iteration 2)

After the first iteration, the system evaluates coverage relative to research dimensions. Missing conceptual areas are identified, leading to new subqueries focused on:

- Long-term societal implications
- Organizational governance strategies

This demonstrates iterative reasoning rather than fixed-depth search.

Step 8 — Memory Reuse and Deduplication

During the second iteration:

- Previously ingested URLs were skipped
- One summary (S3) was reused from memory

This shows efficient knowledge reuse and prevention of duplicate evidence.

Step 9 — Agreement Analysis

Once summaries S1–S4 are collected, pairwise agreement detection is performed.

Observed patterns:

- Strong support between S1 and S2
- Strong support between S3 and S4
- Multiple partial support relationships

Agreement scores increase the strength of corroborated evidence.

Step 10 — Conflict Detection

The system checks for hard factual contradictions between summaries.

Result:

- **No conflicts detected**

This confirms consistency across collected sources.

Step 11 — Evidence-Weighted Scoring

Each summary receives:

- Credibility score
- Agreement score
- Total combined score

S1 achieved the highest total score (18), indicating strong support and reliability.

Step 12 — Report Generation

Validated summaries are passed to the synthesis engine, which produces a structured report with sentence-level citations.

A PDF report is generated automatically as the final output artifact.

5.3 Evidence Summaries

The summaries tab contains dense factual summaries extracted from independent sources. Each summary receives:

- A credibility score
- Agreement-based validation score
- Total evidence strength score

Example observations:

- S1 (score 18) highlights widespread GenAI adoption, productivity gains, and persistent challenges.
- S2 (score 7) provides labor market evidence showing increased hiring associated with GitHub Copilot.
- S3 (score 16) focuses on technical capabilities and ethical/legal concerns of GenAI in coding.
- S4 (score 16) discusses global governance, regulatory fragmentation, and societal risks.

These summaries form the evidence base for synthesis.

5.4 Final Report Generation

The system synthesizes validated summaries into an academic-style research report.

Key characteristics of the generated report:

- Structured sections
- Multi-source synthesis
- Sentence-level citations
- No unsupported claims

A PDF version of the report is automatically generated as the final output of the pipeline.

http://127.0.0.1:7860/files/report_1769743318.pdf

5.5 Automated Evaluation

The evaluation tab provides plan-aware quality assessment.

Evaluation Highlights

- Overall Score: 7.5
- Accuracy: 8 (most claims grounded in summaries)
- Completeness: 6 (some planned dimensions under-covered)
- Citation Quality: 9 (strong traceability)
- Structure: 8 (logical organization)

Identified limitations include incomplete coverage of some research dimensions and a small number of weakly supported statements.

This evaluation demonstrates that the system not only generates reports but also performs self-assessment based on research objectives.

6. Conclusion

This project demonstrates that reliable LLM-based research requires more than advanced language generation — it requires structured reasoning, evidence validation, and controlled synthesis. The AI Research Agent addresses these needs through a modular architecture that separates planning, discovery, reasoning, synthesis, and evaluation into distinct system stages.

Key contributions of the system include iterative research expansion, agreement-based evidence weighting, explicit conflict resolution, persistent research memory, and citation-constrained report generation. Together, these mechanisms transform raw web information into trust-weighted, traceable academic-style reports. Observed system behavior shows improvements in coverage completeness, reduction of logical inconsistencies, and increased grounding through multi-source validation.

However, the system remains dependent on source quality and probabilistic LLM reasoning, and evaluation remains model-based rather than human-reviewed. Therefore, the AI Research Agent should be viewed as a research assistance system rather than a source of verified truth.

Overall, this work illustrates a shift from single-prompt text generation toward **structured research automation**, offering a foundation for future AI systems that prioritize transparency, evidence grounding, and disciplined reasoning.

